

Compiling organizational knowledge into a state an agent can trust

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Abstract

Systems that supply organizational knowledge to language models and agents have no mechanism for telling a consumer that what it retrieved has since become false. When a source document changes, derived artifacts — summaries, extracted obligations, aggregates — become stale silently, and retrieval returns them with undiminished confidence. Recompiling everything on every change is too expensive; recompiling incrementally is unsafe, because the artifact a change *should* have invalidated is the one most likely to be missed.

We describe a knowledge compiler that treats this as a build-integrity problem rather than a retrieval problem: source changes are classified into typed change kinds, propagated along dependency edges typed by change kind, and used to recompile a *candidate* knowledge state which is activated in a single state-transition event, and only after a verification gate passes. We report a pre-registered one-shot controlled holdout in which selective recompilation was artifact-exact equivalent to an independent full rebuild across nine adversarial topologies with zero stale artifacts surviving. A retrospective all-stored regression over 28 natural English-Wikipedia revision pairs also matched the full-rebuild oracle in all 28 pairs with zero stale artifacts left behind; 11 pairs contained semantic changes under the evaluation adapter. Separately, 22 constructed authority, bi-temporal and agent-consumption-lineage cells were each repeated 500 times and all passed — a result we then qualified by ablating the governance filters one at a time, which found six essential, three redundant-but-load-bearing (masked whenever a compliant alternative exists, and the sole refusal when none does), and four not independently measurable by that instrument. We report a pre-registered confirmatory result on a frozen 800-page real-document holdout in which a two-signal selective-acceptance gate was associated with severe-error risk, and we state plainly that its effect magnitude remains uncertain.

We also report what did not work. A structure-only change once left a stale artifact marked current in our own system; that failure is reproduced here because it is the reason one of our mechanisms exists. Our first same-intelligence evaluation was invalidated by leakage and its replacement has not been run: its corpus has since been acquired and verified, and the question set, the leakage measurement and the arms have not. And the semantic identity stage that produces the change set exposed a scalability bottleneck: a sparse shadow matcher removed the measured runtime/memory ceiling, but its eight residual divergences were each a NEW/AMBIGUOUS classification flip, and all eight were measured path-dependent into the following revision. AMBIGUOUS carries no logical identifier, so a flip forks lineage rather than resolving, and that faster matcher was not promoted. We report the performance gain and the promotion veto separately rather than using one to erase the other.

We then built an exact replacement that preserves the full observable matching contract — classification, logical identity, candidate attribution, runner-up and downstream impact — and verified it against mutants that corrupt each of those facets. It is exact and it is two to three times *slower* than the dense baseline it was meant to accelerate. The obstruction is the runner-up rather than the winner: because up to $N-1$ prior units are claimed by other units, a unit's rival can be its $(N+1)$ -th best candidate, so preserving that facet retains

essentially every candidate whenever two revisions have similar unit counts. Exact sparse acceleration of history-sensitive assignment remains open, and we report which facet closes the door. Two claim limitations were also narrowed by their own experiments: a verification-gate ordering limitation that a permutation study could not support, and a ranking element we found provably redundant with another.

1. Introduction

Document-processing systems are usually evaluated as if documents were static: a file arrives, a pipeline extracts structure, and quality is the fidelity of that extraction. Organizational knowledge does not behave that way. Contracts are amended, policies are superseded, specifications are revised, and the artifacts an agent consumes are built from many documents at once. The interesting failure is therefore not a mis-read character. It is an answer that remains confidently available after the document it rests on has changed.

This turns document processing into a compilation problem over a mutable world. A revision arrives; some semantic units continue their prior identity and some do not; artifacts that depended on the changed units become stale; the rest may be carried forward; and the resulting candidate state may be activated only if it can be shown to contain no stale artifact. Each of those steps is a place where a system can look successful and be wrong.

Three properties make the problem harder than incremental build systems, which solved a superficially similar problem decades ago. First, **identity is not given**: a build system is told which file changed, whereas a revised document must have its units matched to their predecessors, and that matching can be ambiguous. Second, **the dependency edges are typed by change kind**: a reordering that alters no text still invalidates artifacts aggregated over document position, and a formulation that seeds invalidation only from content changes will silently carry those forward — we measured exactly this failure. Third, **abstention is a required outcome**: where identity cannot be settled, the honest result is an unresolved marker that forces a rebuild, not a confident guess that rewrites a history.

This paper reports what happens when those properties are taken seriously and then measured rather than asserted. Several of our results are negative, and they are the ones we consider most useful: a faster identity policy that we vetoed, an exact replacement for it that turned out slower than the baseline it was meant to accelerate, and a claim limitation we withdrew after a permutation study failed to support it.

A note on what this system is not. It is not a better OCR engine and not a retrieval-augmented generation pipeline. Extraction quality and retrieval are inputs to the problem studied here, not the problem. Nothing in this paper claims an improvement to either.

Contributions.

1. A change-kind-typed dependency formulation in which a change that alters only document *structure*, producing no content-level change, still seeds invalidation — motivated by a measured failure of the naive formulation.
2. A pre-activation verification gate — a conjunction of candidate-state predicates whose acceptance result we measured to be invariant to evaluation order — under which an artifact whose input fingerprint cannot be verified is treated as *unverifiable rather than passing*, and a candidate state containing one is not activated.
3. A pre-registered one-shot controlled holdout establishing artifact-exact equivalence between selective recompilation and independent full rebuild.
4. A retrospective real-corpus regression over all 28 already-stored natural Wikipedia revision pairs, reported explicitly as non-confirmatory and single-source-family evidence.
5. A characterisation of where the approach's cost actually sits, including a scalability bottleneck we did not initially expect and do not conceal.
6. A set of negative and invalidated results retained rather than discarded, including a faster identity policy that failed a later-revision invariance criterion and therefore remained shadow-only.

7. An exact-semantics replacement for that policy, verified against mutants corrupting each facet of the matching contract, which is **slower** than the dense baseline it was built to accelerate — together with a post-hoc structural account, that preserving the runner-up rather than the winner is what removed the sparsity advantage **under this certification strategy**.
8. Two claim limitations narrowed by experiments designed to test them: a verification-gate ordering limitation that a permutation study could not support, and a ranking element found provably redundant with another.

Contributions 5 through 8 are negative or narrowing results. We list them as contributions rather than as caveats because each changed a decision — one vetoed a promotion, one closed a design direction, and two removed claim language.

1.1 Related work

Incremental and selective rebuild. Dependency-driven partial rebuild is old and well understood, from `make` [1] through modern build systems organised around content addressing [2, 3], and through data-lineage tooling in the analytics world [4, 5]. We claim no novelty in incremental rebuild as such. What differs here is that the dependency edges must be *typed by the kind of change* that traversed them, and that the nodes' identities are inferred rather than declared. A build system is told that `a.c` changed; we must first decide whether this revision's paragraph is the same paragraph as last revision's.

Entity resolution and record linkage. Matching units across revisions is a one-to-one assignment problem, and maximum-weight bipartite matching is the standard tool [6, 7]. The record-linkage literature [8, 9] focuses on match quality; our constraint is different and stricter. Because a match decides which *history* continues, an assignment that is equally good by score but different in attachment is not equivalent for our purposes — it forks a lineage. Section 4.9 reports what that constraint costs when one tries to accelerate the matcher.

Bitemporal data and as-of queries. Valid-time and transaction-time modelling is prior art [10, 11] and we use it as such. It appears here only composed with authority ordering and permission filtering, and §4.6 reports an ablation showing which of those filters are individually load-bearing and which are redundant-but-still-load-bearing when no compliant alternative exists.

Retrieval-augmented generation. RAG [12] addresses which passages to place in a context window. It does not address whether the corpus behind those passages is currently consistent, nor what must be rebuilt when it is not. The two are complementary, and we make no comparative claim against retrieval systems.

Transactional publication. Our activation step is an *ordered refusal followed by a single-event switch*, and the property a consumer observes is that no intermediate state is visible [13, 14]. It is not a distributed transaction protocol and we do not evaluate crash behaviour, partial-failure recovery or multi-node consensus. §4.7 states the boundary explicitly because it is the easiest of our results to over-read.

2. Method

Figure 1 shows the whole path a revision takes, from source document to a consumer's context window, and marks the two places the system is allowed to refuse: the identity resolver's abstention and the activation gate. The four subsections below take those stages in order.

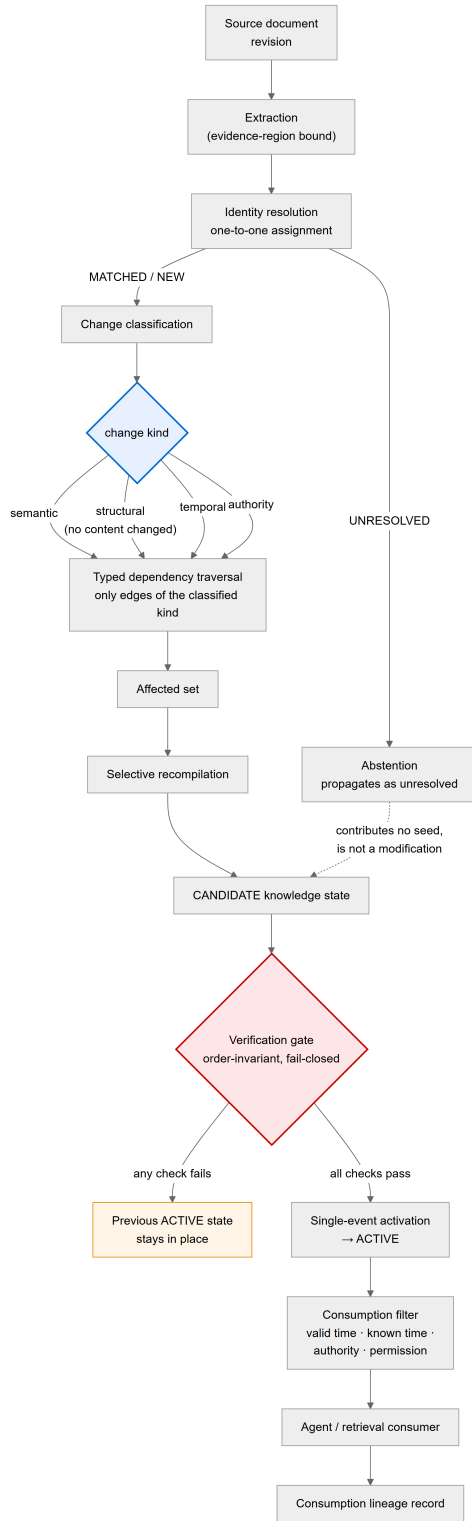


Figure 1. Where staleness enters, and where it is stopped

2.1 Stable semantic identity

Each derived knowledge unit carries a logical identifier intended to survive source revision. Resolution between versions is a one-to-one assignment: no prior unit may be claimed as the continuation of two incoming units, because independent per-unit resolution silently forks histories. Scoring is a weighted mean over seven signals renormalised over the signals that *have values* — a missing signal is recorded with a reason, never filled with zero — and when a signal designated critical has no value the resolver abstains. Figure 4 gives the outcome

space of a single resolution — matched, new, or unresolved — and shows that an unresolved unit contributes no traversal seed rather than a default one.

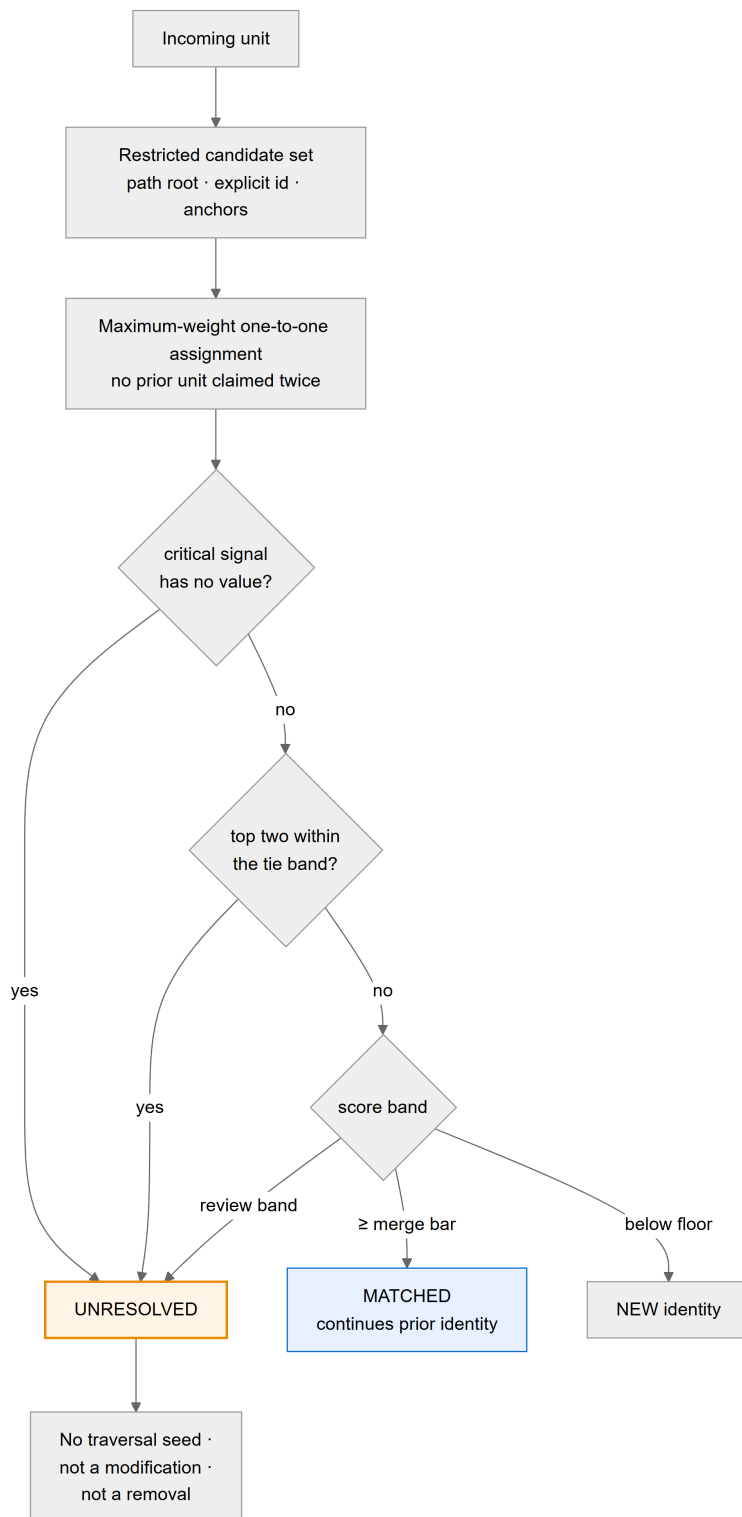


Figure 4. Identity resolution outcomes

The bands are deliberately conservative: continuation at ≥ 0.92 , a reviewable band down to 0.75, a new identity below that, and abstention when the two best candidates fall within 0.05 of each other. The asymmetry is intentional. A false merge rewrites the history of a clause and no later query can detect it; a false split is visible and joinable. **These thresholds are not calibrated.** The system's own calibration table refuses to

report itself calibrated without a named corpus, and we do not present them as measured optima.

2.2 Typed change classification and impact

Differences between versions are classified into semantic, structural, temporal and authority change kinds. Dependency edges are typed by change kind, and a derived artifact declares which kinds it is sensitive to. Impact is computed by traversing, from the logical identifiers associated with each classified change kind, only the edges typed to that kind.

The structural channel exists because of a failure, described in §4.4.

2.3 Verification before activation

Selective recompilation produces a *candidate* knowledge state. Before it can become active, a gate checks that every classified change is accounted for by the affected set; that every affected artifact was rebuilt; that each carried-forward artifact's recorded input fingerprint — covering declared inputs *and* structural facts — matches; and that no artifact whose fingerprint cannot be verified is marked current. An artifact that declares a structural dependency without declared structural coverage is unverifiable, not passing. Activation occurs in a single state transition, conditional on all checks, with no partially activated state visible to a consumer. Figure 3 draws the four checks in the order the implementation evaluates them; §4.8 reports that the order is not load-bearing.

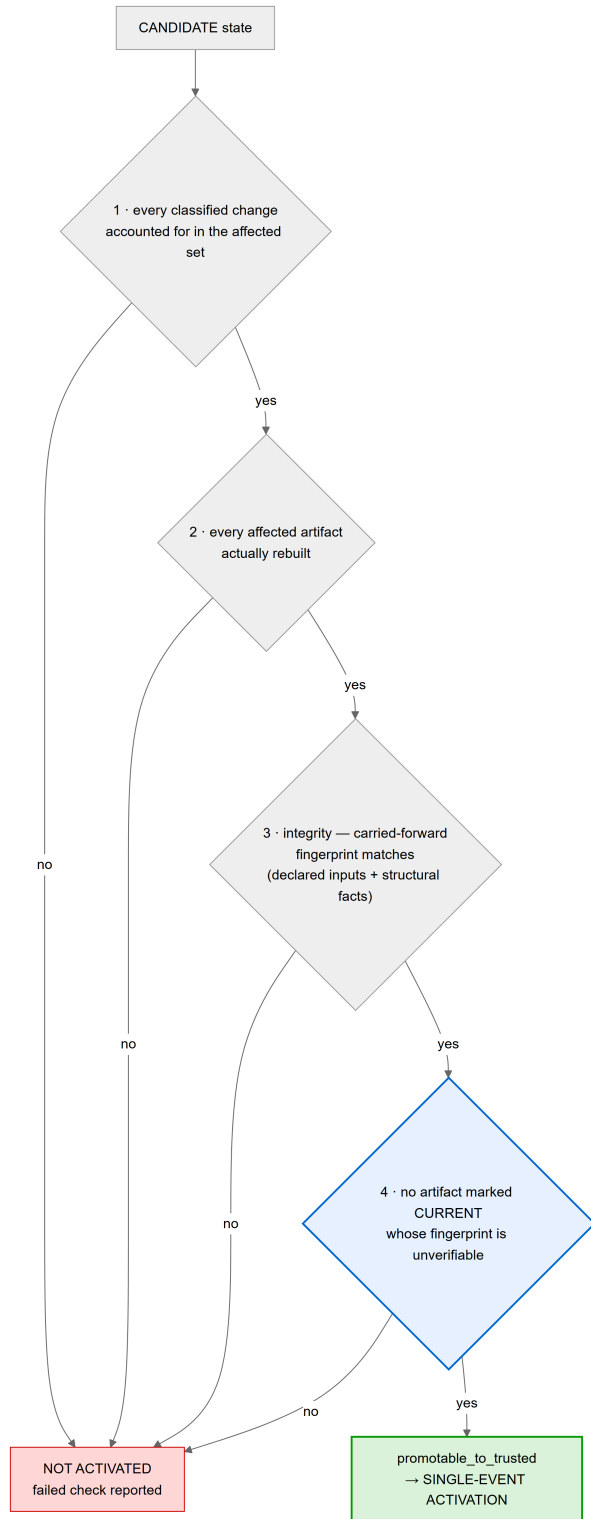


Figure 3. The gate's four checks, drawn in their implementation order

What the build receipt does and does not establish. The receipt seals `artifact_id` $\leftrightarrow$ `artifact_sha256` $\leftrightarrow$ `input_fingerprint` $\leftrightarrow$ `build_action_id` $\leftrightarrow$ `parent_world_state_id` $\leftrightarrow$ `builder_version`. It establishes that what was declared was what was built. It does **not** establish that the declaration was complete: a wrong input set can be sealed perfectly. We state this explicitly because the seal's cryptographic strength invites exactly the wrong inference.

2.4 Retrieval under world, time, authority and permission

Queries are answered against the active state under a composed filter: valid time and known time, authority and applicability, and permission. A unit excluded by the permission filter is excluded such that its identifier is not revealed — an excluded-but-named unit is a leak.

3. Experimental design

3.1 Pre-registration and one-shot discipline

Protocols are frozen before data are seen, and thresholds are never widened after a result. Where a protocol had to change, the change is recorded as a numbered amendment stating what changed, why, and whether any criterion moved; in every amendment reported here, no acceptance criterion moved.

The Family B holdout is **one-shot**. Its generator version, seed hash, success criteria and code were frozen before it was run, it was run once, and it has not been re-run or re-tuned. A further change to Family B requires a new seed and a new protocol, not a second attempt at this one.

3.2 Controlled versus real

We distinguish three evidence classes throughout and never let them substitute for one another:

class	meaning	used for
controlled	generated corpus with known ground truth	mechanism correctness
real	external document or revision corpus	external validity
implementation	unit tests only	neither

Controlled evidence establishes that a mechanism does what it claims on inputs we constructed. It says nothing about how often those inputs occur in practice. Synthetic revisions are used for capability testing and are never substituted for real temporal evidence.

4. Results

4.1 Selective recompilation equals full rebuild (controlled, confirmatory)

On a pre-registered one-shot fresh-seed holdout, selective recompilation was artifact-exact equivalent to an independent full rebuild in every cell, with zero stale artifacts surviving and zero escapes missed by the gate.

cell	cases	equivalent	escaped artifacts
unchanged	400	400	0
semantic_only	400	400	0
structural_only	400	400	0
mixed	400	400	0
ambiguous	400	400	0
deep_chain_64	200	200	0
cross_document_dependency	200	200	0
diamond_conflicting_channels_semantic	200	200	0
diamond_conflicting_channels_structural	200	200	0

gate_missed_escape_total = 0; primary_success = true.

Reported against ourselves: the semantic_only and ambiguous cells rebuilt more than strictly changed — 400 and 454 false invalidations respectively. Over-rebuilding is the safe direction, but it is a cost and it is not netted

out of the saving reported in §4.3.

4.1.1 Natural-revision regression (real corpus, retrospective)

We then ran the **current pinned implementation** over every natural revision pair already stored across the Family B development, confirmatory-corpus and extension stores: 28 English-Wikipedia pairs / 56 source files. This protocol was frozen to include *all* stored pairs rather than selecting titles by outcome, but the data had been used by earlier work, so this is a retrospective regression — not a fresh holdout and not a confirmatory endpoint.

Selective state matched the independent full-rebuild oracle on **28/28 pairs**, with **zero stale artifacts left behind**. Eleven pairs contained semantic changed logical identifiers under the evaluation adapter; the other 17 normalized to no semantic change. The mean planned rebuild fraction across all 28 pairs was 7.94%.

This closes a narrower question than the controlled holdout: current code also reproduces equivalence on the natural revision material already in the repository. It does **not** establish general equivalence. English Wikipedia is one source family, the dependency graph is an evaluation adapter, and SEC/DART/contracts/ scans and unconstrained production graphs remain outside this result.

4.2 Authority, temporal integrity and lineage (controlled)

Across **22 constructed mechanism cells**, each repeated 500 times (11,000 executions), every cell passed. The resolver selected the correct authority, abstained where an override lacked evidence, marked exactly-equal conflicts conflicted rather than choosing, never revealed a permission-hidden claim identifier, reproduced as-of answers on both temporal axes, and reconstructed consumption lineage to depth 64 at precision and recall 1.0 including against decoys.

The limit of this result, stated as the receipt states it: controlled mechanism correctness only; no automatic temporal extraction and no real-world frequency or generalization claim. In particular, valid-time intervals are supplied correctly by construction. *Reading* a validity interval off a document is a different capability and is untested. This result must not be written as "the system understands temporal validity from documents."

4.3 Cost, split into the two stages it actually has

Downstream of change detection, given an already-computed dirty set, selective recompilation was **3.62× (1,000 units)** and **4.31× (10,000 units)** faster than full rebuild, at a planned-rebuild fraction of ~1.1% and ~1.0%.

The identity and change-detection stage that *produces* that dirty set scales far worse. Figure 5 puts the two stages side by side, which is the comparison a single end-to-end speedup number would hide:

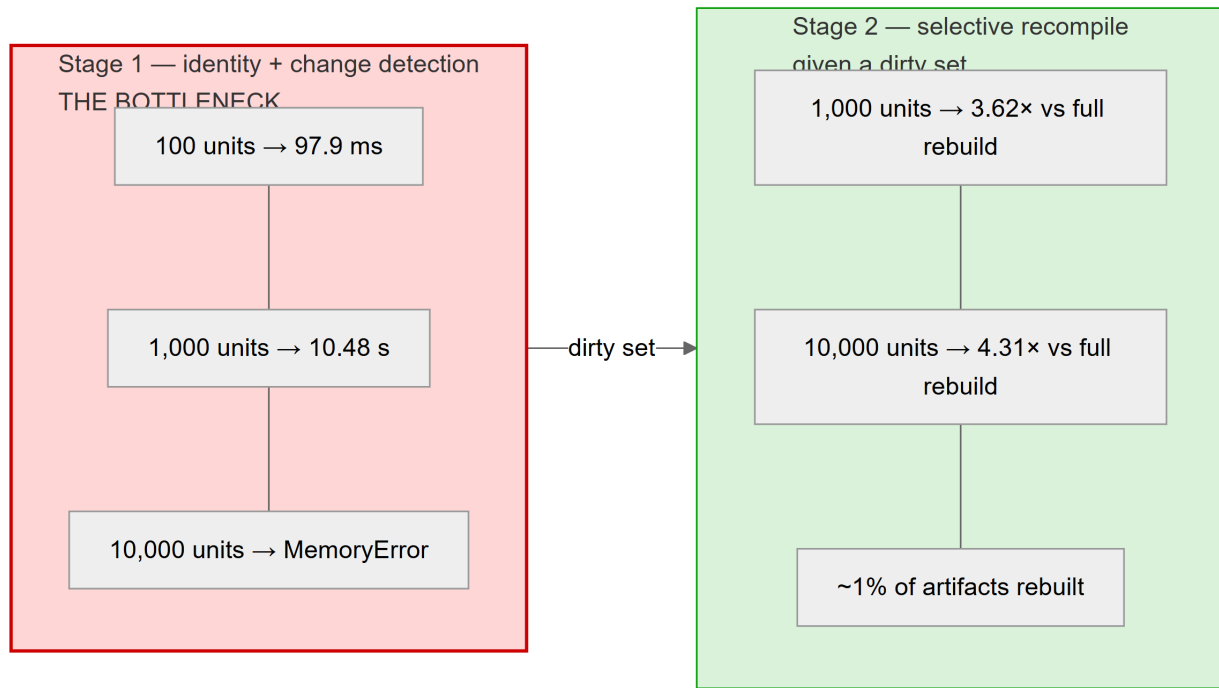


Figure 5. Cost, split into the two stages it actually has

units	identity + change detection, p50
100	97.9 ms
250	1.20 s
500	3.02 s
750	6.01 s
1,000	10.48 s
10,000	MemoryError

These must never be combined into one end-to-end speedup. Doing so would hide the dominant cost inside the incremental saving, and it is the first objection a reviewer should raise. We raise it ourselves.

The MemoryError above is a historical measurement from before an exact memory fix that changed the assignment matrix from (float, dict, dict) cells to floats and recomputed signal detail only for assigned pairs. That fix moved the LEGACY ceiling: subsequent probes completed 1,500 and 2,500 units, so the table must not be read as the current 10,000-unit failure boundary.

A candidate-indexed BLOCKED shadow policy went much further in a separate assign_one_to_one microbenchmark: 767 ms at 1,000 units versus 36.0 s for the unrestricted matcher (46.97× in that specific measurement), and 12.9 s at 10,000 units with a 33 MB peak. Those timings are **not comparable** to the identity-plus-change-detection table above and are not an end-to-end speedup.

More importantly, performance did not determine promotion. Across 33,600 controlled identity decisions the shadow policy produced zero measured fail-open and zero fail-closed divergences but eight residual tied split assignments. A follow-on protocol frozen before its future-state endpoint found that **all 8/8** row-attached divergences produced a different next-revision decision; the negative control produced zero differences and the positive falsification arm separated. The current BLOCKED implementation is therefore PROMOTION_VETO_PATH_DEPENDENT and LEGACY remains default. The speed result is retained; exact Protected-Core equivalence is not claimed.

4.4 A negative result we caused: structure-only staleness

An earlier version classified a change that altered only the order of units as producing no changed logical identifiers. The impact traversal therefore had an empty seed, and a position-aggregated artifact — genuinely stale — was carried forward as CURRENT. The promotion gate blocked the run.

We record two things about this that are easy to get wrong. First, **the block is not a pass**: equivalence had already been violated by the time the gate fired, and scoring the run as a success because the safety net held would be scoring the wrong thing. Second, the system never published a wrong world state; what happened is that the planner produced a false-negative classification and the first version of the gate had a fingerprint blind spot. The structural dependency channel and the extended fingerprint exist because of this. Figure 2 puts the defective path and the corrected one side by side; it is the clearest statement of what the typed-dependency channel adds over a content-hash incremental build.

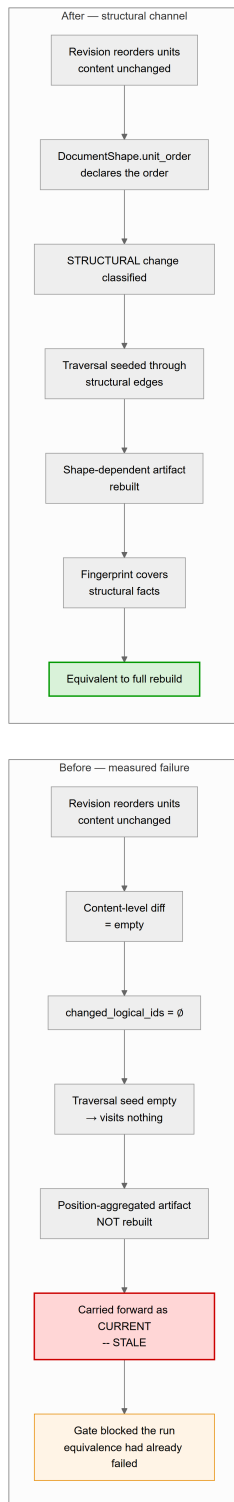


Figure 2. The failure this system caused, and the fix

4.5 Selective acceptance on real documents (confirmatory)

The pre-registered primary test showed an association between the combined selective-acceptance gate and severe-error risk on the frozen holdout (Fisher exact $p = .032$), while the corrected risk-difference bootstrap interval included zero, so the magnitude of risk reduction remains uncertain.

749 of 800 holdout pages carried an official score. Accepted: 648 pages, 38 severe. Abstained: 101 pages, 12 severe. Bootstrap interval for the risk difference: $[-0.21 \text{ pp}, +12.79 \text{ pp}]$.

Disclosure required by our own rules: the first evaluator used for this endpoint was non-deterministic. The result above is the corrected run. The flawed run and the comparison between them are retained in the repository rather than replaced. This is a post-unsealing correction and is disclosed as one.

A cohort limitation, not a design choice: no source-grounded arm was possible, because the frozen cohort is scan-image centred and carries no compatible native text layer to ground against.

4.6 Same intelligence, better world: NOT RUN

The primary endpoint of our comparison between raw text, basic retrieval and a compiled world has **not been run**, and we report the reasons rather than a weakened substitute.

The first question set was invalidated by leakage: Recall@1 of 0.9969 with a same-document share of 1.0 meant the baseline was not a baseline. The second was built correctly but produced zero revision-sensitive facts, because the available 56-pair corpus consisted of consecutive revisions in which infobox facts do not move. We declined both available escapes — lowering the threshold, and manufacturing questions from prose diffs — and declared the endpoint not run. A candidate manifest of 6,987 long-horizon titles was frozen *before* any revision outcome was inspected, so no selection leakage has been introduced; acquisition against it failed twice on API transport rather than on science.

A third acquisition succeeded and was spent on development, deliberately. It returned 750 titles, and a protocol conflict was found in the run that used them. Rather than publish that run as an endpoint, we sealed those 750 titles as development data. They are excluded from the confirmatory candidate pool by a fail-closed guard, leaving **6,237 untouched titles**. We report the guard's own failure too: an earlier version read a key the seal does not carry, so it removed nothing while reporting success. An empty development-title list is now a refusal, because an inert guard is worse than no guard — it produces a receipt saying the check passed.

The confirmatory apparatus is built, frozen and gated; the endpoint is still not run at the time of writing. What is fixed before any holdout number exists:

- *The model is pinned by attestation, not by name.* Two runtimes serving the same name can differ in checkpoint, tokenizer, quantization and serving stack. The candidate registry's expected revision, the revision attested from the live runtime, and the revision in the config freeze are the same string, and the checkpoint's files were verified against the digests the model hub declares for that commit, with zero mismatches. An unattestable runtime fails the gate; there is no fallback to another model.
- *Raw text receives the whole document pair*, not a shared retrieval budget. Under a shared 3,000-token budget the development run measured raw text losing the answer to truncation on 0.5763 of questions — that is a context-window effect, and reporting it as a representation effect would have been the easiest way to overstate our result. Exceeding the real context window is recorded as an overflow outcome, never truncated away silently.
- *One arm-fairness metric is demoted rather than used.* The share of questions separable between the two revisions is 1.0 by construction of the admission rule. We declare it a generator integrity check and do not cite it as evidence that the task is hard.
- *The model's default reasoning preamble is disabled in every arm.* **What was measured**, on the live runtime and reproduced on a second one: with the default chat template this checkpoint had not reached an answer at our fixed 256-token limit, and with the preamble disabled the same runtime returned a bare value in six completion tokens and the abstention sentinel in seven. **What we inferred from that**, and state as inference: under the same budget an enabled preamble would truncate mid-reasoning in every arm, so the run would report the preamble rather than the four representations. The setting is applied identically to all four arms, which changes the shared model contract and not the variable under test, and it means our result may not be read as measuring this model with reasoning enabled.

The gate that stands in front of the holdout has 21 conditions, all currently passing, and each one is falsified in the same execution to show it can refuse.

The confirmatory run was executed on 2026-08-20 and stopped before its arms. The holdout was acquired, a question set was generated from it, three pre-arm gates were measured, one refused, and no arm was run. We report that outcome rather than the result we set out to obtain.

Acquisition. The holdout title list is the frozen candidate manifest with the sealed development titles removed: 6,237 titles, overlap zero. A first acquisition attempt spent two and a half hours in rate-limit backoff and then died on a deterministic path error in its own invocation --- the same defect this repository had already diagnosed once and recorded --- writing no receipt, no question and no result. Its partial output is quarantined unexamined and was not reused. The second attempt ran over the identical sealed manifest, seal and cutoffs; its command line was read from the operating system and the failing expression simulated for both attempts before it was allowed to continue. It returned **391 eligible revision pairs and 359 named exclusions from 750 titles consumed, with none unaccounted for**, and a corpus root hash over every verified pair. Nine fail-closed integrity checks passed, including zero overlap with the sealed development titles.

Question set. The frozen generator produced **2,548 questions: 185 revision-sensitive primary and 2,363 unchanged controls**, over 265 documents with parsable infoboxes.

Gates. The structural taint gate passed --- the reference builder produced zero tainted queries and all three mutants were caught at 477 of 477. The residual leakage gate passed --- residual overlap median and 90th percentile both 0.0 against frozen ceilings of 0.05, with the injected positive control exceeding and the clean control within. **The baseline validity gate refused.**

The refusal, exactly. Six of its seven criteria passed. One did not:

The v8 development set passed the frozen tie-rate criterion (7.34%), whereas the untouched confirmatory holdout exceeded the predeclared 10% ceiling (19/166, 11.45%).

A tie at the top of the ranking is resolved by index order, so a tied question is a ranking decision the retriever did not make. Because the rate is a ratio of integers over 166 evaluated questions, the ceiling has an exact integer form: 16 ties is the largest passing count (9.64%), 17 already fails (10.24%). **The holdout exceeded the largest admissible count by three questions.** We do not present this as a marginal 1.45-percentage-point excess; the criterion is a deterministic comparison over counts, and $19 > 16$.

The ceiling's provenance, because it is the obvious place to attack this result. The v7 corpus measured 10.38%, which is **above** 10% --- the historical observation motivated having the criterion, and v7 would not have passed it. Ten percent was adopted as a round operational-quality ceiling, deliberately stricter than the observation rather than fitted to it. It was pinned in a hashed configuration freeze at 02:50:43Z, **28 minutes before the holdout title manifest existed** and six hours before the holdout question set. It was not relaxed after the holdout was measured. We do not claim more than that: **10% is a predeclared operational validity threshold, not a theoretically derived constant.**

What this result is. The confirmatory benchmark failed its pre-arm retrieval-validity criterion, so no representation-comparison endpoint was run. The comparison between raw text, basic retrieval, enriched retrieval and a compiled world is **not measured** --- not lost, not tied, not shown to be absent. No arm ran; no answer was ever generated by the model; the measured inference cost of this endpoint is zero.

One diagnostic does survive, and it is not about our system. Under the frozen baseline retrieval representation, the strongest superseded-revision unit outscored the oracle current-revision unit for **64.46%** of evaluated primary questions (median gap -0.0537). We label this a **baseline retrieval difficulty observation**: it says temporal retrieval on this corpus is hard for the frozen baseline. It is not evidence for our system, and no arm result exists that could make it so.

What we did not do. We did not raise or remove the ceiling, regenerate the question set with different admission or tie-breaking, delete questions, change the retriever, run the arms anyway, or build a replacement benchmark. The threshold was fixed before the cohort existed and the cohort has now been measured against

it; any adjustment made after seeing 11.45% would convert a confirmatory run into a development one and consume the only untouched cohort this programme has. Those titles are development data from this point forward.

We consider a not-run endpoint with an intact protocol more useful than a run endpoint with a compromised one.

4.7 The activation gate refuses, enumerably (contract audit)

The verification-and-activation path of §2.3 was audited under a sealed protocol: the live `world_state` and `build_receipt` modules were hashed and pinned, a list of 21 required contract behaviours was fixed, and the suite was executed. All 21 behaviours were present and 49 tests passed.

The behaviours are enumerated rather than summarised, because "the gate works" is not a result. Publication is refused on: a checksum mismatch; an integrity failure; a permission check that was not executed; a missing or failing equivalence check; an unstaged object; a receipt whose seal has been broken or whose fields were tampered with; a receipt bound to a prior world or to another artifact; wrong artifact bytes; and moved inputs. A failed publish leaves the previously ACTIVE state in place and emits **no** event; a successful publish emits **exactly one** event, bound to the manifest hash. A rejected candidate cannot be retried and the same candidate cannot be published twice. Rollback is supported, and refused for unpublished or current states. A builder without tracked capture cannot reach HIGH_INTEGRITY; the permissive path yields only DEGRADED_UNTRACKED.

What this is, stated exactly. This is a controlled implementation and unit-contract audit. It is *not* a demonstration of distributed atomicity. We have not exercised a process crash mid-transaction, a partial database, network or storage failure, or multi-node concurrent publication. "Atomic" here means the conditional single-step activation of a verified candidate, not a distributed transaction guarantee.

It also does not move the build-record trust boundary. A builder that never records an input class cannot be detected by receipt binding alone; the only guarantee available is fail-closed or degraded treatment of untracked builders. Reporting tests-passing as an empirical result about production behaviour would be exactly the promotion this paper's status vocabulary exists to prevent.

4.8 The gate's check ordering is commutative, and the claim was narrowed

The promotion gate evaluates four checks in a fixed order, and an earlier draft described that order as part of what prevents a stale artifact from being promoted. Nothing measured it.

We permuted all 24 orders of the four checks over six constructed candidate states (144 evaluations), including the case where an ordering effect would be most likely to appear: a state that is internally consistent **and** incomplete. **No permutation changed the admit/refuse verdict.**

A null is only as good as the instrument's ability to have returned something else, and ours initially could not. The first harness was voided by a fidelity gate that requires the canonical order to reproduce the real gate's verdict and its first failing step; it failed on a state the real gate refuses. The second harness passed fidelity but implemented the four checks as pure functions of the candidate state, so no permutation could have diverged whatever the truth was -- the same non-separating defect we had already recorded for an earlier falsification arm. Only after adding a positive control, a deliberately order-dependent gate on which the harness diverged for 12 of 24 orders, did the null become interpretable.

The result is that the gate's acceptance decision is a conjunction of candidate-state predicates and is **invariant to evaluation order**. We initially wrote that the four checks are *mutually independent*, and that was an overclaim: order-invariance does not entail independence, since a conjunction of coupled predicates is still commutative. A static audit of the implementation found a coupling the permutation study could not have detected -- where no full-rebuild oracle is available, the integrity check's pass is discharged by the staleness check that follows it, and the source says so. The checks are order-invariant; they are not independent, and we

report the weaker proposition because it is the one the evidence carries.

The ordering determines which refusal is reported, not whether one occurs. We narrowed the corresponding patent limitation accordingly, under a rule frozen before the experiment ran.

4.9 Exclusivity is what makes identity expensive, and it does not prune

Our identity resolver assigns incoming units to prior units by a global maximum-weight matching, so one prior unit cannot be claimed as the continuation of two new ones. That exclusivity is the property that keeps histories from forking silently, and §4.3 reports the cost: a dense $N \times M$ scoring pass.

We first pursued the obvious acceleration -- restrict each unit's candidates to a structurally plausible bucket. It is fast (~47x at 1,000 units) and it is wrong: 8 residual decisions diverged, and all 8 propagated into the following revision as lineage forks. It remains shadow-only under a promotion veto. Figure 7 traces how a window-restricted matcher turns a contended assignment into a fork, and Figure 8 tabulates the resulting trade with the negative result kept in the table rather than dropped from it.

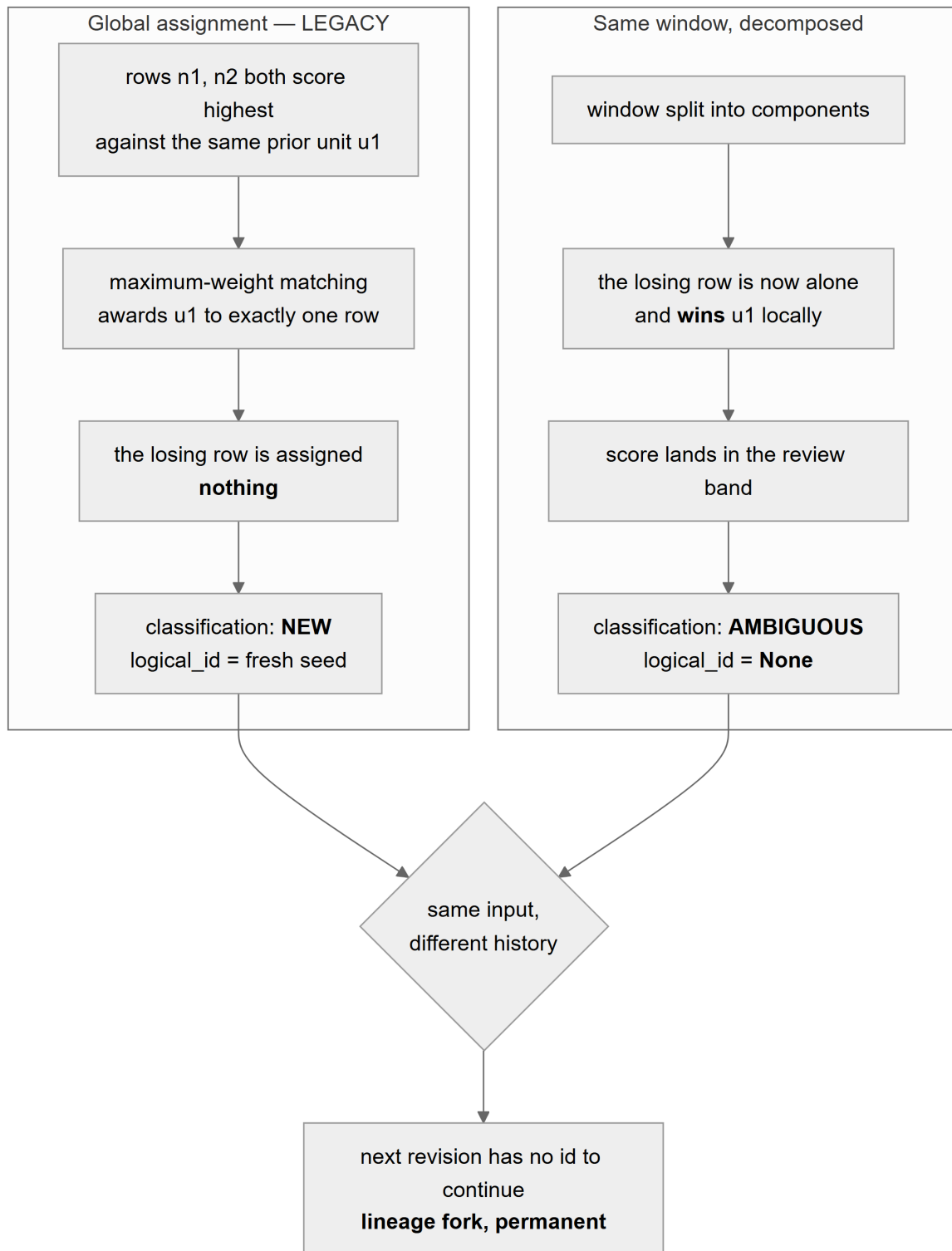


Figure 7. why accelerating identity matching broke history

Figure 8 — the identity scalability trade, including the negative result

policy	speed	exactness	status
LEGACY (dense global)	baseline	exact by definition	in production
BLOCKED (bucketed candidates)	~47x at 1,000 units	8 residual, 8/8 future-path divergence	shadow only — promotion veto
CERTIFIED_SPARSE (admissible bound + certificate)	0.35–0.48x	exact on frozen corpus, 3 separating mutants	not proposed for promotion

Figure 8. the identity scalability trade, including the negative result

We then asked *why*, and the mechanism is sharper than "approximation loses accuracy". Under the global assignment, a unit that loses a contended prior unit to another unit is assigned nothing and is classified NEW. Split the window -- which is what any decomposition does -- and the same unit wins that prior unit locally and lands in the ambiguous band instead. Since an ambiguous decision carries no logical id, the next revision seeds a fresh identity, and a single flipped classification becomes a permanent fork. Exclusivity is a *global* property, and locality destroys it.

Finally we built the exact version. It never removes a candidate from the problem; it computes a cheap admissible upper bound and pays for an exact score only where the bound cannot certify the pair irrelevant. It reproduces the full observable contract with zero divergences across 50 comparisons under candidate-order and row-order perturbation, verified against three mutants that each corrupt one of those facets.

"Full" is meant literally and was checked. All twelve facets of the frozen contract are compared directly and all are exact: classification, logical id, assigned column, candidate attribution, relation, score, unresolved seeds, dependency impact, next-revision lineage, the serialized change record, the changed-identifier set, and the selective rebuild set. Four of those were unmeasured until we audited our own contract against our own harness, and the route to that list is itself reported in §5.1 -- including the fact that an earlier version of this sentence claimed twelve while enumerating eleven.

Twelve facets is not twelve results. All of them are compared over the *same* constructed corpus of nineteen cases, and two of those cases were added after the first runs failed to reach the code paths under test. Adding facets to a comparison can only reveal divergence and never conceal it, so each additional facet strengthens the conclusion — but none of them widens the population it holds over. The one arm that is corpus-independent is the replay against the identity-scalability generator at a pinned seed, 70 comparisons with 0 divergences, and we report it separately rather than folding it in.

It is 2-3x slower than the dense baseline. The obstruction is the runner-up, not the winner. Because up to $N-1$ prior units are taken by other rows, a unit's runner-up can be its $(N+1)$ -th best candidate, so preserving that facet requires retaining $\min(M, N+1)$ candidates per unit -- every candidate, when the two revisions have similar unit counts.

We also ran it against the bucketing policy's own adversarial topologies, driven from the same generators at the same pinned seed, and it is exact on all seven cells including the two -- split-and-merge and near-tie -- that produced the eight divergences which vetoed the fast policy. So the exactness is not an artifact of a gentle corpus: it holds precisely where the fast policy failed. It is still slower.

We report this as a negative result, and state its scope precisely: **under the tested certification strategy, preserving global assignment, candidate attribution and runner-up semantics removes the expected sparsity advantage.** That is a statement about one strategy on one corpus, not about the problem. Exact sparse acceleration of history-sensitive assignment remains open; a different certificate for the runner-up, or a formulation that does not require preserving it, is untested here and is not excluded by anything we measured.

4.10 An abstention says which signal was missing, and why (contract audit)

§2.1's identity resolver refuses rather than guesses when a signal it treats as critical has no value. A refusal that says only "ambiguous" is indistinguishable from a bug, so the resolver records what it had. We audited that record under the same sealed protocol as §4.7: the live identity module was hashed and pinned, a list of 12 required contract behaviours was fixed, and the suite was executed. All 12 were present and passed.

The recorded behaviours are, again, enumerated rather than summarised. An abstention caused by an absent signal carries the identities of the signals that *were* scored, the identities of those that were **not**, and for each absent one a reason drawn from the four the implementation distinguishes: the model was unavailable, the signal was not applicable to this pair, a resource limit was reached, or the evaluation errored. The two maps are disjoint, so an absent signal is never given a numeric value — a zero is indistinguishable from measured disagreement, and the two demand different recoveries. The operator-facing reason names the absent critical signal. The whole record survives serialization.

Two controls decide whether that means anything. The positive control is a pair on which every signal has an input: it must produce an empty absence map and score all seven weighted signals, or the recital would be satisfied vacuously by reporting absence unconditionally. The break controls construct the two defects the contract forbids — a signal recorded as absent *and* scored, and a free-text absence reason — and require the same predicates the passing tests rely on to reject both. A guard that cannot fail is not a guard, and this programme has found that defect in its own detectors often enough to test for it by default.

What this is, stated exactly. Controlled implementation evidence over the live module. It establishes what the implementation records when a signal has no value. It does **not** measure how often absence occurs in real documents, and it must not be promoted to a real-data result. It also asserts no completeness: the four reasons are what this implementation distinguishes, and the receipt records `absence_taxonomy_completeness_claimed: false`. The corresponding patent claim was amended the same day so that it recites an *enumerated reason distinguished by the implementation* rather than "the reason each absent signal had no value", which would have read onto a complete taxonomy nobody has.

Finally, "distinguishable from a failure to evaluate" is established in the narrow sense the tests exercise. An abstention carrying an absence record is distinguishable from a *tie* abstention, which is reached after a complete evaluation and therefore carries an empty absence map; and an evaluation that is refused outright produces no decision object at all, so there is nothing to mistake for either. This is not a crash-safety or partial-failure result.

4.11 What the element-support pass changed about the claims

The evidence in this paper is also the support for a patent application, and the two are audited against each other element by element rather than claim by claim, because an examiner objects to a limitation and not to a claim. That audit reported eight limitations whose recited scope reached past the receipt beneath it. We record here what was done about them, because the alternative — quietly leaving the broader wording in place — is the failure mode this whole programme is built to avoid.

Six were closed by **amending the recital down to the measurement**. The selective-acceptance holdout of §4.5 records attribution at page granularity and compares per-page critical-token correspondence, so the two independent-claim limitations that had recited *per-unit evidence regions* now recite what the receipt contains. The change-classification limitation had listed four change kinds in one recital, and only two of them — semantic and structural — are exercised over the real revision corpus of §4.1; the other two are exercised over constructed cells, and now sit in a dependent claim at that class. The dependency-graph limitation had recited that a derived artifact authors its own sensitivity declaration, which no receipt isolates from the typed graph itself. The identity limitation had bound the measured half — identifier persistence across revisions — to an unmeasured half in the same sentence. And the unresolved-identity limitation had recited a three-part negative behaviour of which one part is observable in the ambiguous holdout cell.

One was closed by **producing evidence**: §4.10 above.

One was closed by **removing the recital from the independent claim entirely**. It recited admission of accepted representations to a downstream trusted store. No receipt exercises that, and the acceptance outcome has no consumer in the live system outside its own module and unit tests. The available options were to build an admission path so the limitation could be audited, or to move the limitation to a dependent claim held for a continuation. We took the second. Building a mechanism in order to make a claim testable inverts the relationship between the system and the evidence, and a reader who later found that the admission path existed only because a claim needed it would be right to distrust everything else.

The honest accounting of that trade. Removing a limitation from an independent claim makes it easier to infringe and harder to allow. Eleven of the twenty-four registered amendments are therefore recorded as broadenings, five as narrowings, and the direction of each is *derived from the operation* by a fixed table rather than asserted by its author — a control that exists because an earlier pass of this work recorded four broadenings as narrowings and the mis-accounting hid where support was owed. Every removed recital is

preserved in a dependent claim and in the specification. None is discarded, and none is described as a narrowing.

5. Limitations and threats to validity

1. **Controlled corpora still dominate.** §4.1 and §4.2 establish mechanism correctness on generated inputs. §4.1.1 adds a 28-pair retrospective natural Wikipedia regression, but it is one source family and does not establish that controlled change mixes or conflict patterns occur at these rates in practice.
2. **No threshold here is calibrated.** The identity bands (0.92 / 0.75 / 0.05) are conservative choices, not measured optima.
3. **Consumption completeness is unevidenced.** The build receipt cannot establish that the declared input set was complete. This is an open trust boundary and its long-term closure is not started.
- 3a. **Activation is contract-audited, not crash-tested.** §4.7 is a controlled unit-contract audit. Process crash mid-transaction, partial storage/network failure and multi-node concurrent publication are not demonstrated.
4. **Effect magnitude in §4.5 is uncertain.** The interval contains zero.
5. **External validity is narrow.** Natural-revision equivalence is now measured on one English-Wikipedia source family. Unseen domains, genuine amendment relationships in SEC/DART, unseen layouts and language diversity remain untested.
6. **Valid-time extraction is untested**, as distinct from valid-time reasoning.
7. **Identity scalability and identity equivalence are not simultaneously closed.** The sparse shadow removes the measured scaling ceiling, but its eight residual divergences are not permutations between tied halves: each is a **classification flip** between NEW and AMBIGUOUS, and all eight were measured path-dependent into the following revision. AMBIGUOUS carries no logical identifier, so a flip forks lineage at the next revision rather than resolving. The shadow is not promoted, and LEGACY remains the correctness baseline.
8. **Single-host timings.** All cost figures are from one CPU host; no distributed or contended measurement exists.
9. **One evaluator defect was found after unsealing** (§4.5) and corrected. Others may exist.
10. **The same-intelligence comparison is NOT_MEASURED.** §4.6's confirmatory run stopped at a pre-arm validity gate, so no arm was executed and no comparison between raw text, basic retrieval, enriched retrieval and a compiled world exists. This is not a null result and must not be read as one: nothing was measured that could have come out either way.
11. **The tie ceiling is operational, not derived.** The 10% exact-tie ceiling that stopped §4.6 was a predeclared operational validity threshold, not a theoretically derived constant. It was frozen 28 minutes before the holdout title list existed and was not relaxed afterwards, but we do not claim more for it than that.
12. **One holdout cohort, one source family, retrospective revisions.** §4.6 used English Wikipedia revision pairs separated by at least 1,095 days. There is no cross-source natural-revision validation, and the cohort is now spent: those titles are development data and cannot serve as a holdout again.
13. **Three reproducibility defects in §4.6's own instrumentation are disclosed rather than repaired in place.** The baseline gate's receipt carries a hard-coded `run_class: DEVELOPMENT_ONLY` although its recorded input digest is the holdout question set's (W6-V8-GATE-01); the endpoint sealer's arm-receipt filename map was written before the gates ran and silently omitted two receipts, corrected by an append-only supplement rather than by editing a sealing tool after its outcome was known (W6-V8-SEAL-01); and one intermediate version of the instrumentation provenance receipt was edited in place and is lost, so we claim neither full write-once history nor that every version was preserved.
14. **§4.6's first acquisition attempt failed on its own invocation**, not on science: 149 rate-limit retries followed by a deterministic path error that this repository had already diagnosed once. Its partial output is quarantined unexamined and was not reused; the second attempt ran over the identical sealed manifest,

seal and cutoffs.

15. **Two receipts do not self-verify under the manifest's rule.** Both v8 question-set receipts hash every field except the question list, by their generator's own convention, while the experiment manifest recomputes over the whole document. Each is internally consistent under its own rule and inconsistent under the other. The generator is frozen and digest-pinned and the holdout is closed, so it was not edited to resolve the mismatch after the fact.
16. **§4.10 is a contract audit, not a measurement.** It records what the live implementation does when an identity signal has no value. It does not measure how often that happens on real documents, and it asserts no completeness over the absence taxonomy: the four enumerated reasons are what this implementation distinguishes, and the receipt says so in a field rather than in prose.
17. **Four claim recitals are supported only by the implementation, and are held rather than filed.** Per-unit evidence-region binding, downstream trusted-store admission, the B-family evidence-region binding, and artifact-authored sensitivity declaration each exist in code with unit tests and are exercised by no receipt. They are recited in reserved dependent claims, kept out of the filing core, and reported in the element-support matrix at severity DISCLOSED_RESERVED_NOT_IN_FILING_CORE. None was made testable by building a mechanism for it.
18. **Three claims have no recoverable amendment history.** The claim set is deliberately untracked in version control, and the hash chain that replaces version history was pinned partway through the programme. It detects change from that point forward and reconstructs nothing before it, so for three dependent claims the record cannot distinguish "never amended" from "amended without an entry". They are reported UNDETERMINED, never "unamended".
19. **A withdrawn assertion survived in a registry field no detector scanned.** Three paper_section labels in the claim-evidence registry used the "atomic promotion" wording narrowed on 2026-08-19. The assertion audit scans surfaces — claim set, specification, manuscript, figures, tables — and a registry field that no surface printed was outside that set. It fired the moment Appendix A began rendering the field. Corrected; the general limitation stands, because every registry field is a candidate for future rendering and none of the others has been swept.
20. **One full-suite run recorded three failures that did not reproduce, and their identities were not captured.** The run used a summary-only traceback setting; a re-run under the same interpreter and the same collection (3,102 items) reported 3,034 passed, 68 skipped, zero failed. We record the discrepancy rather than resolving it, because we cannot: the failing test names are unrecoverable, and asserting a cause — contention, ordering, or anything else — would be a reconstruction, not a finding.

5.0.1 Selection and leakage

The 28 natural revision pairs are **every pair already stored** at the time of the regression, not a sample drawn for it. No pair was selected, excluded or reordered after any outcome was visible, so there is no selection leakage to inherit.

All-stored is not the same as randomly sampled. The pairs were originally acquired for a different purpose, and whatever biases governed that acquisition are inherited here unmeasured. We claim the absence of *outcome-dependent* selection, which is a narrower and checkable property, and not representativeness.

The 6,987-title manifest for the unrun retrieval experiment was likewise frozen before any revision outcome was inspected, so a future retry inherits no selection leakage either — a property that survives the acquisition having failed twice.

5.1 Failure analysis

Every failure below is retained in full in the repository's failed-and-superseded ledger. We summarise them here because a results section that omits them would misrepresent how the system reached its current form.

A structure-only revision left a stale artifact current. A revision that reordered units and changed no text produced no changed logical identifiers, so the impact traversal started from an empty seed and a position-aggregated artifact was carried forward as current. Equivalence was violated before the promotion gate fired. We record this as a failure and not as a success of the safety net: the gate holding is not evidence that the planner was right.

A confirmatory evaluator was non-deterministic, and the defect was found after unsealing. The corrected figures are the citable ones and the correction is disclosed as post-unsealing rather than presented as the original result. The associated magnitude claim was withdrawn entirely: the risk-difference interval contains zero, so the association survives and the effect size does not.

A retrieval baseline was invalidated by leakage. Questions built from the retrieved passage produced recall near 1.0. A binding upper-bound gate caught it, which is the argument for making ceilings binding rather than advisory.

A falsification arm could not falsify. Its weakening could not change any answer by construction, so its zero divergences tested nothing. We have since hit the same class of defect twice more — a permutation arm that could not diverge, and an exactness corpus that could not reach the code paths under test — and the rule now applied is that a null is uncitable until a positive control on the same instrument separates.

A faster identity policy was vetoed. Discussed in §4.9. It remains shadow-only.

An exact replacement for it was slower than the baseline. Also §4.9. We report it because identifying *which* facet defeated this strategy is more useful than another unexplained negative -- while noting that this identifies the obstruction for the certificate we tested, not for the problem.

We asserted, in our own specification, that an optimisation was exact when our own experiment said otherwise. The specification described the component decomposition as exact; it is exact only against the already-restricted candidate graph, and the measured divergence against the unrestricted problem is the reason the policy is shadow-only. The claim text had been narrowed when the result landed; the specification had not been swept in the same pass, and neither had a figure caption. The rule now applied is that a narrowing updates claims, specification, manuscript and figures together or it is not finished.

We described a contract facet as preserved when nothing had measured it. The exact matcher's frozen contract lists next-revision lineage as an observable facet; the harness stopped at the current revision, and a summary table said the facet was preserved anyway. It turned out to be true when we measured it, and that is not a defence: a sentence written from what a mechanism should do rather than from what a receipt says is an overclaim whatever its truth value. Two further facets were then found unmeasured and argued "strongly implied" -- the same reasoning again -- so they were measured too. The sentence that then announced all twelve facets covered was *itself* unsupported: it enumerated eleven. The missing one, the assigned column, was the single per-row facet the comparison function never read, and it had been called indirectly implied by the logical identifier -- which does not follow, because that identifier is read from the matched partner, so two prior units sharing one are indistinguishable through it. We measured it directly by observing the resolver's own entry points rather than reconstructing the assignment: zero divergences over nineteen cases, with instrument liveness recorded alongside the null. Four occurrences of one habit, the last of them inside the correction to the third.

We destroyed one of our own receipts. A tooling change intended to redirect an output applied incompletely, and a re-run overwrote an earlier readiness receipt while reporting it had written elsewhere. The path now holds a deterministic regeneration whose hash necessarily differs from the lost original; both hashes are pinned and the incident is retained. We report it because a rule against destroying evidence is worth nothing if a violation of it can be tidied away.

Our self-audit has not demonstrated convergence, and we decline to imply that it has. Three successive adversarial passes were run over this evidence base, each begun after the previous one had been declared complete. Each found new material defects: the first, two statements in our own specification that our own receipts disprove; the second, a facet described as preserved that nothing had measured; the third, a failure whose *cause* we had inferred from a disk signature and never verified, a scope sentence that claimed twelve items while listing eleven, and a "repository green" label attached to a subset of the test suite. The rate of discovery did not fall.

The natural inference — that a fourth pass would find more — is one we accept rather than argue against, and a fourth pass duly found three more: two limitations we had withdrawn from claims on evidence still asserted in five other places, a set of exactness results resting on a single corpus, and a suite of test figures that described a different environment than the one they were run against.

We have since frozen a stopping rule rather than continuing to decide after each pass that enough had been done. It requires two consecutive rounds of eight heterogeneous detectors — each with a stated blind spot, each carrying a positive control that must fire in the same execution — with every prior finding closed. **Zero such rounds have been completed.** The first was invalidated because its human pass recorded no findings by an argument rather than a reading, and the argument was one we constructed after noticing it would make the round clean. What we claim is narrower than an absence of defects: every finding is recorded with the receipt that closed it, no finding was closed by weakening the wording it contradicted, and the failed and superseded artifacts are retained rather than repaired. A reader should treat the audit as a process with a known nonzero yield, not as a certificate.

We think the shape of these defects is the transferable part. All of them are the same act: a sentence written from what a mechanism *should* do rather than from what a receipt *says*. Three were about our own mechanisms and one was about our own failure, and the last of the four was produced by the fix for the third. The mitigations that now exist — binding prose numbers to live receipts, requiring a positive control before a null is citable, recording instrument liveness beside a zero — were each added after that act had already happened once.

5.2 Summary tables

The full table set is in docs/paper/TABLES_2026-08-19.md. Five tables carry the evidence base: the experiment hierarchy with statistical units separated from repetition counts (Table 1), selective-recompilation equivalence (Table 2), the three-policy identity comparison (Table 3), the ablation results (Table 4), and the blocked/not-run inventory with its unblock contracts (Table 5).

Figure 6 is the companion to those tables: it places every claim in the set on its evidence class, including the NOT RUN and BLOCKED ones, so that the shape of what is unproven is visible at the same glance as what is.

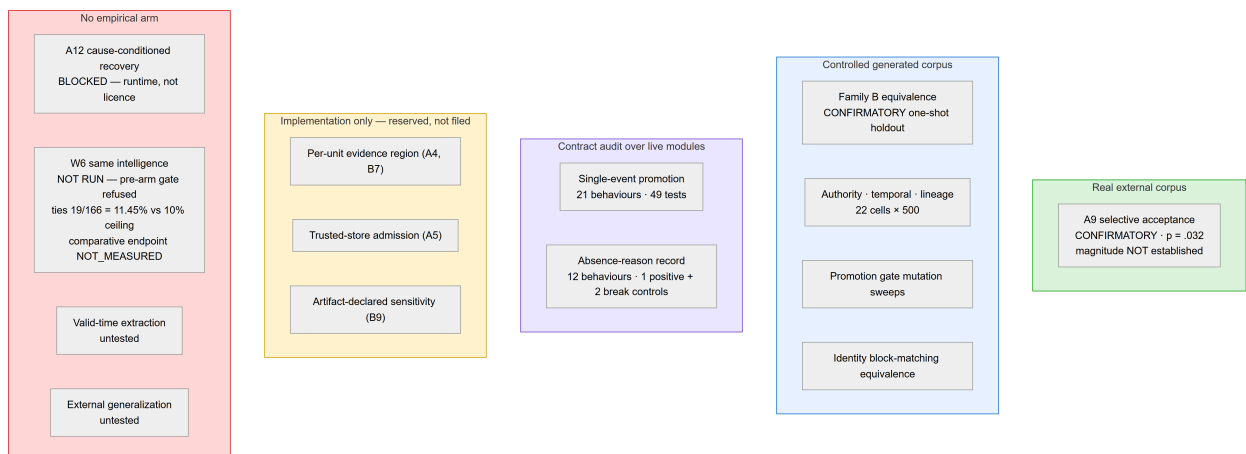


Figure 6. Evidence classes across the claim set

Every table separates `n_units` from `n_repetitions`, because a constructed cell repeated 500 times is 500 repetitions of one unit and not 500 samples of the world. Table 3 is the one we would ask a reader to look at first: it puts a fast policy that produces a different history and an exact policy that is slower than its own baseline side by side, which is the honest state of the identity work.

6. Reproducibility

Every result cites a receipt carrying its own sha256, the sha256 of the protocol that governed it, the sha256 of the code modules it exercised, and its seed. Superseded receipts are retained and are pinned by hash from the receipt that supersedes them; none is edited or deleted. The claim–evidence matrix is generated by a tool that re-reads every asserted number out of its receipt by JSON pointer and refuses to emit if any disagree, and that checks the pinned source modules still hash to what the measurement recorded.

At the time of writing that inventory is 24 experiments and 181 receipts, with zero undeclared code-pin drift and **two receipts whose self-recorded hash does not recompute**. We name them rather than round them away: both are v8 question-set receipts, and the cause is a convention mismatch, not tampering. Their generator deliberately hashes every field *except* the question list, while the manifest recomputes over the whole document minus the hash field. Each receipt is internally consistent under its own rule and inconsistent under the manifest's. The generator is a frozen, digest-pinned source and the holdout it served is closed, so it is not edited to resolve this after the fact; the discrepancy is disclosed here and in the reproducibility limitations instead. Two receipts *do* carry declared drift: the identity module was reformatted after they were measured. We did not edit them to remove the mismatch. They are listed in a drift register that states which is superseded and which is claim-carrying, and the claim that cites the latter states the drift in its own text; the manifest generator fails on undeclared drift only, so the gate stays meaningful without rewriting history.

Three honest qualifications about this section, because a reproducibility claim is the easiest one to overstate:

- The package digest is taken at **manifest time, not experiment time**. It is not a lockfile and we do not describe it as one.
- Of 24 experiments, exactly **one** has its original command line recoverable from a receipt. That number is about *history* and does not improve: the other 23 invocations are gone. Separately, **eight** experiments now have an exact current-session command recorded, because their scripts take no arguments and the invocation is fully determined; the remainder carry reproduction *templates*. The execution index keeps these three categories apart rather than presenting a reconstructed command as history.
- Two artifacts in the repository are not reproducible in principle: a readiness receipt we destroyed with our own tooling, whose replacement is a deterministic regeneration with a necessarily different hash, and a partial acquisition directory left by a defect that our own diagnosis initially misattributed to a dying process. Both are retained with their status recorded.
- **The interpreter is part of a result, and we learned this the expensive way.** Every test figure this programme published was produced by an interpreter that is not the project environment. Under it, 24 tests failed in a way that reads as a product defect: the document-parsing sandbox launches its child process in isolated mode, which excludes user-installed packages, and the dependencies were installed there. The child could not import them, exited non-zero, and every downstream assertion saw an unfinished analysis task. Under the project environment the same isolated import succeeds. Nothing in the product was wrong, and nothing in the sandbox was relaxed to discover it — isolation is the boundary that keeps a hostile document away from the host, and trading it for a green number was not available. Build status is now emitted by a tool that records the interpreter and withholds the word *green* when it is not the project one.

7. Discussion

7.1 What the results support, at what strength

Three claims are carried by pre-registered one-shot evidence, and they are the only three: selective recompilation reproduces an independent full rebuild artifact-exactly on an adversarial controlled holdout; the activation gate refuses a candidate state containing an artifact whose inputs cannot be verified; and a joint agreement-and-stability gate over two extraction configurations is *associated* with severe-error risk on a frozen 800-page real-document holdout.

The third is the weakest of the three and the most likely to be over-read. It is an association at $p = .032$. The bootstrap interval for the risk difference contains zero, so **no magnitude is established**, and no sentence in this paper says the gate reduces severe errors by any amount.

Everything else is controlled-corpus mechanism evidence, contract-audit evidence over live modules, or implementation only — and the paper's figure 6 draws the four classes as four boxes rather than merging them into "validated".

7.2 What the negative results changed

This programme's negatives did more work than its positives, and each changed something concrete rather than being noted and set aside.

The **structure-only staleness defect** is the reason the change classification carries a structural kind at all. It was produced by an earlier version of this system: a change that altered only document order produced no changed logical identifiers, left the traversal seed empty, and carried a stale position-aggregated artifact forward as CURRENT. The limitation that fixes it is now the most distinctive element of the independent claim, and it exists because the obvious design — classify by content — misses exactly the case with no content delta.

The **identity path-dependence probe** vetoed a promotion. The candidate-indexed matcher is roughly $47\times$ faster and diverged on 8 of 33,600 decisions; all 8 diverged again in the following revision, and an AMBIGUOUS classification carries no logical identifier, so each flip forks lineage. A speedup that rewrites history is not a speedup, and the policy remains shadow-only.

The **exact certified sparse matcher** removed the last route around that: it preserved the full contract and ran at $0.35\text{--}0.48\times$ the dense baseline. We report that as a property of one certification strategy on one corpus. It bounds that strategy and it bounds nothing about the problem: a different certificate for the runner-up, or a formulation that does not require preserving it, is untested here and is excluded by nothing we measured.

The **governance filter ablation** is the one that cost claim scope. Permission, temporal-validity and applicability filters turned out to be redundant for the verdict whenever a compliant alternative exists, and the sole refusal only when none does. Four ranking elements were not measurable at all by that instrument. The claim may not be prosecuted as though each filter were independently necessary, and it is not.

The **gate-ordering ablation** narrowed an element to an order-invariant conjunction, and the specification took a year of drafting inertia to catch up: two sentences still asserted that the order was material until 2026-08-20.

7.3 The comparative endpoint, and why its absence is load-bearing

The experiment this programme most wanted was the simplest to state: give one fixed model a compiled world versus raw text versus basic retrieval, and see whether the compiled world answers revision-sensitive questions better.

It did not run. The confirmatory holdout failed a predeclared pre-arm retrieval-validity criterion — exact ties at $19/166 = 11.45\%$ against a 10% ceiling frozen six hours before the question set existed — and the arms were not executed. **The comparative endpoint is NOT_MEASURED**, which is a different statement from a comparison that ran and came out flat. Measured inference cost: \$0.00.

The honest consequence is that this paper establishes *mechanism* properties of a knowledge compiler and establishes **nothing** about whether a model answers better with one. A reader who wants that number will not find it here, and the cohort that would have produced it is spent: it is development data now, and a future

comparative result requires a fresh untouched cohort at real cost.

We report the stop rather than the workaround because a gate honoured only when the excess feels small is not a gate. The excess here was three questions.

7.4 What we would tell someone building this

Three things, in the order they cost us most.

Separate operational failure from semantic failure, and separate absence from zero. Both are the same discipline applied at different layers: a signal with no value is not a signal scoring zero, and a pod that died is not a model that was wrong. Every place this programme collapsed the two, it produced a number that looked like evidence and was not.

Build the detector's positive control before the detector. Every audit in this repository runs a probe through its own classifier in the same execution, because a guard that cannot fire reads identically to a clean result. We learned this by shipping guards that could not fire.

Assume a withdrawn claim survives somewhere. Three separate incidents (P19, P22, P29) found a retracted proposition alive in a document that a sweep had declared clean, each time in a form the previous detector was not built for. Nothing here establishes that the current detectors catch the next one, and the useful posture is to expect the fourth rather than to declare the class closed.

7.5 Threats we cannot design around

The controlled corpora still dominate. The natural-revision regression is 28 pairs from one source family, acquired for another purpose, and it is *every pair already stored* rather than a sample — which buys the absence of outcome-dependent selection and buys nothing about representativeness. No threshold in this system is calibrated; the identity bands are conservative choices, not measured optima, and `CalibrationTable.calibrated` is `False` and refuses to be set true without naming a corpus.

And the build receipt cannot establish that a builder's declared input set was complete. That is a boundary of the mechanism rather than a gap in the measurement: a wrong input set can be sealed perfectly, and the only guarantee available is fail-closed or degraded treatment of untracked builders.

8. Conclusion

We treated document processing as compilation over a mutable world and measured what that costs.

Selective recompilation reproduced an independent full rebuild artifact-exactly on a pre-registered controlled holdout, and on all 28 natural revision pairs we had already stored — the latter retrospective, single-source-family, and explicitly not a generalisation to arbitrary documents. The verification gate refuses a candidate state containing an artifact whose inputs cannot be verified, and treats unverifiable as distinct from passing; a permutation study showed its acceptance decision is invariant to check ordering, so we narrowed the corresponding claim rather than defend an ordering we could not show mattered.

The results we expect to be most useful are the negative ones. A candidate- bucketing identity policy is roughly forty-seven times faster and produces a different history, so it is vetoed. An exact replacement preserving the full observable contract is slower than the dense baseline it was meant to accelerate, because preserving which unit would have been the *runner-up* requires retaining essentially every candidate when two revisions have similar unit counts. Exclusivity is what makes inferred identity trustworthy, and it is also what makes it expensive; we can now say which facet closes the door rather than reporting an unexplained failure.

Three questions stay open and we mark them as open rather than answered: whether exact acceleration is achievable under a cheaper runner-up certificate; whether these results hold on a second source family, for which no corpus with explicit amendment relationships was available to us; and whether the thresholds throughout are calibrated, which they are not and which no result here claims.

9. References

Citations in this paper are to background work only. Nothing in this list is used as a comparison baseline, and no result here is measured against a number reported in any of these works — §7.3 states why the comparative endpoint is absent rather than borrowed. The evidence for this paper's own claims is cited by artifact and hash in Appendix A, not here.

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Appendix A — evidence classification of every claim

The full matrix, with every asserted number re-read from its receipt by JSON pointer, is `docs/ip/CLAIM_EVIDENCE_MATRIX.md` (generated; it refuses to build if any claim cites a receipt that is missing, unhashable, or does not say what the claim says it says). The index below is the same registry summarised, so a reader with only this manuscript can see the shape of the evidence.

Status vocabulary is deliberately finer than "supported". CONFIRMATORY is pre-registered and one-shot; EMPIRICALLY_VALIDATED_CONTROLLED is a generated corpus; EMPIRICALLY_VALIDATED_REAL is a real external corpus; IMPLEMENTED_ONLY is code and unit tests with no experiment; NOT_RUN is a valid protocol whose endpoint

never executed; FAILED and SUPERSEDED are retained, never deleted.

claim id	status	receipts bound	paper section
A9-SELECTIVE-ACCEPTANCE-ASSOCIATION	CONFIRMATORY	1	A. Recovery Runtime / Selective Acceptance
A9-EVALUATOR-NONDETERMINISM	SUPERSEDED	2	Limitations / threats to validity
A2-ABSENCE-REASON-CONTRACT	IMPLEMENTED_ONLY	3	C. Living Knowledge — identity
A12-CAUSE-CONDITIONED-RECOVERY	BLOCKED_INTERNAL	5	A. Recovery Runtime — stated as not run
B-EQUIV-FRESH-HOLDOUT	CONFIRMATORY	2	C. Living Knowledge — incremental recompilation
B-EQUIV-REAL-CORPUS-RETROSPECTIVE	EMPIRICALLY_VALIDATED_REAL	1	C. Living Knowledge — real-corpus regression
B-STRUCTURE-ONLY-STALE-DEFECT	FAILED	2	C. Living Knowledge — negative results
B-PROMOTION-GATE-STALE-BLOCK	EMPIRICALLY_VALIDATED_CONTROLLED	3	C. Living Knowledge — single-event world-state activation
B-SAFETY-B2-CONSUMPTION-COMPLETENESS	PARTIALLY_SUPPORTED	1	Limitations / threats to validity
B-PERFORMANCE-SPLIT	EMPIRICALLY_VALIDATED_CONTROLLED	1	C. Living Knowledge — cost
B-IDENTITY-SCALABILITY-ROOT-CASE	IMPLEMENTED_ONLY	1	C. Living Knowledge — identity
B-IDENTITY-BLOCKED-EQUIVALENCE	PARTIALLY_SUPPORTED	5	C. Living Knowledge — identity
B-IDENTITY-SCALABILITY-MEASURED	EMPIRICALLY_VALIDATED_CONTROLLED	1	C. Living Knowledge — identity
B-ATOMIC-PROMOTION-CONTRACT	EMPIRICALLY_VALIDATED_CONTROLLED	3	C. Living Knowledge — single-event world-state activation
B-EQUIV-CROSS-SOURCE	NOT_RUN	1	C. Living Knowledge - external validity, stated as not run
D-GOVERNANCE-LINEAGE-CONTROLLED	EMPIRICALLY_VALIDATED_CONTROLLED	2	C. Living Knowledge — authority, temporal, lineage
W6-SAME-INTELLIGENCE	NOT_RUN	26	D. Same Intelligence, Better World — reported as not run
SCOPE-NARROWING	IMPLEMENTED_ONLY	1	Related work
B-GATE-ORDERING-NOT-LOAD-BEARING	EMPIRICALLY_VALIDATED_CONTROLLED	8	B. Single-event activation / Gate composition
B-GOVERNANCE-FILTER-ABLATION	PARTIALLY_SUPPORTED	4	B. Governance / authority resolution
B-LEGACY-MATCHING-INVARIANTS	EXPLORATORY	7	B. Identity resolution / scalability
B-EXACT-SPARSE-UNRESOLVED	FAILED	9	B. Identity resolution / scalability

Receipt paths and sha256 digests for every row are in the generated matrix and in docs/repro/EXPERIMENT_MANIFEST.json (24 experiments, 181 receipts, zero undeclared code-pin drift, two receipts whose self-recorded hash does not recompute — see §6 and limitation 15).

Appendix B — retained failed and superseded results

See docs/evidence/FAILED_AND_SUPERSEDED_LEDGER.md.

Appendix C — tables

Reproduced in full from docs/paper/TABLES_2026-08-19.md, the file the manuscript cites. Table numbering is that file's.

Every table below separates the **statistical unit** from the **repetition count**. A generated corpus repeated 500 times is 500 repetitions of one constructed cell, not 500 independent samples of the world, and presenting it as the latter is the commonest way an evidence base is silently inflated.

Column meanings, fixed for every table:

- `n_units` — independent constructed or natural units. This is the number that bounds what can be generalised.
- `n_repetitions` — executions per unit. Bounds only variance, never scope.
- `evidence_class` — one of controlled, real-retrospective, contract-audit, blocked, not-run.

Table 1 — Evidence hierarchy across the programme

#	experiment	n_units	n_repetitions	evidence_class	outcome
1	Selective-vs-full equivalence, fresh holdout	9 adversarial topologies	pre-registered one-shot	controlled	artifact-exact equivalence, 0 stale
2	Selective-vs-full, natural corpus	28 revision pairs (11 with semantic change)	1	real-retrospective , one source family	28/28 matched oracle, 0 stale
3	Governance / bitemporal / lineage	22 constructed cells	500	controlled	22/22 pass with all filters enabled
4	Governance per-filter ablation	13 filters × paired+solitary states	1	controlled	6 essential, 3 redundant-but-load-bearing, 4 unmeasurable
5	Single-event promotion contract	21 enumerated refusals	49 tests	contract-audit	21/21; <i>not</i> distributed atomicity
6	Gate-ordering ablation	6 constructed states × 24 orders	1	controlled	144 evaluations, 0 verdict divergences
7	Identity: dense vs candidate-indexed	7 topologies, 33,600 decisions	1	controlled	8 divergences , each a NEW/AMBIGUOUS classification flip, all path-dependent (a flip forks lineage)
8	Identity: path-dependence probe	the 8 residual divergences	1	controlled	8/8 diverge in the next revision
9	Identity: exact certified sparse	19 cases + 7 topologies	order perturbations	controlled	0 divergences , and 2–3× slower
10	Selective-acceptance gate	749 scored pages of an 800-page frozen holdout	1	real , pre-registered	association $p = .032$; magnitude interval contains zero
11	Same-intelligence retrieval	391 eligible pages, 293 revision-sensitive attributes across 151 articles	0	not-run	corpus acquired and verified ; endpoint not executed
12	Cross-source equivalence	—	—	not-run	no second source family available

#	experiment	n_units	n_repetitions	evidence_class	outcome
13	Cause-conditioned recovery	48-case source bundle ready	0	blocked	no runtime image digest obtainable
14	Absence-reason record (identity signals)	12 required contract behaviours over the live module	1	contract-audit	12/12 with one positive and two break controls; <i>not</i> a real-data measurement

Reading rule. Rows 1, 3, 4, 6, 7, 8 and 9 are constructed. Only rows 2 and 10 involve real documents, and row 2 is a single source family examined retrospectively. No row licenses a statement about arbitrary documents.

Table 2 — Selective recompilation equivalence

corpus	n_units	equivalent to full rebuild	stale artifacts surviving	rebuild fraction	evidence_class
fresh adversarial holdout	9 topologies	all	0	—	controlled
natural revision pairs	28 pairs	28 / 28	0	mean $\approx$ 7.94%	real-retrospective

Of the 28 natural pairs, **11** contained semantic changes under the evaluation adapter; the remaining 17 exercise the no-change path. The mean rebuild fraction is computed over the pairs that changed.

What this table is not. It is not a benchmark against another system, and it is not a sample of arbitrary documents. It is English Wikipedia, retrospective, all pairs already stored — no fresh holdout was drawn.

Table 3 — Identity matcher comparison

The central table of the identity story.

policy	semantic exactness	future-path exactness	candidate attribution	runtime vs dense	promotion status
LEGACY (dense global assignment)	exact by definition	exact by definition	exact by definition	1.0× (baseline)	in production
BLOCKED (candidate-indexed buckets)	8 divergences / 33,600 decisions	0/8 preserved — all 8 diverge next revision	not preserved	~47× faster at 1,000 units	shadow only — promotion veto
CERTIFIED_SPARSE (admissible bound + certificate)	0 divergences (50 comparisons + 70 topology comparisons)	0 divergences at revision 2 (19 cases, measured)	preserved	0.35–0.48× , i.e. slower	not proposed for promotion

The two failures are different in kind. BLOCKED is fast and produces a different history. CERTIFIED_SPARSE produces the right history and is slower than the thing it was meant to accelerate.

Why CERTIFIED_SPARSE is slower — structural, not an optimisation defect. Preserving the runner-up requires retaining $\min(M, N+1)$ candidates per unit, because up to $N-1$ prior units are claimed by other units. When two revisions have similar unit counts, $N+1 > M$ and every candidate is retained.

A note on how this table was checked. Its future-path column originally read "preserved on tested corpus" while nothing had measured next-revision lineage — the exactness harness stopped at revision 1. That was an overclaim, found in our own second red-team pass and closed by running the measurement rather than softening the wording.

That pass then found three further contract facets uncomparred, and the sentence announcing all twelve covered turned out to enumerate **eleven**: the assigned column, the one per-row facet the comparison function

never read, was absent and had been called "indirectly implied" by the logical identifier. It is not — the identifier is read from the matched partner, so two prior units sharing one are indistinguishable through it. Measured directly by observing the resolver's own entry points: **0 divergences over 19 cases**, with instrument liveness recorded beside the null (72 observations, 67 assigned columns). All twelve facets are now compared directly.

How many results this row is. One. The exactness figure is a single measurement over a single constructed corpus of 19 cases, examined along twelve contract facets; the facet extensions are additional *comparisons* over the same corpus, not additional corpora. Adding facets can only reveal divergence and never hide it, so they strengthen the conclusion — but they do not widen the population it holds over, and two of the 19 cases were added after the first runs failed to separate. The 70 H1-E topology comparisons are the one corpus-independent arm, drawn from a different generator at a pinned seed, and are reported separately for that reason.

The $\min(M, N+1)$ account is post-hoc. It is a derivation from the matching formulation rather than a pre-registered prediction, and it was written after the timing result it explains. It is offered as a structural explanation, not as a tested hypothesis.

Not established. That exact acceleration is impossible; that a maximum-weight assignment is theoretically required; that no faster exact algorithm exists. What was tested is one certified strategy, and it did not accelerate.

Table 4 — Ablations: which mechanisms are load-bearing

mechanism	ablation	result	claim consequence
structural change channel	mechanism absent (reproduced defect)	stale artifact carried forward as current	claim retained; it is why the mechanism exists
promotion gate integrity checks	declaration + build-record mutation	gate refuses, 21 enumerated cases	claim retained
gate ordering	all 24 permutations	0 verdict changes	claim narrowed to an order-invariant conjunction
candidate index safety	candidate set emptied	80 divergent decisions vs 0 intact	comparator demonstrably separates
tie guard / critical-signal abstention	H1-E2 path-dependence probe	8/8 path-dependent, each a classification flip that forks lineage	vetoed a promotion
permission / temporal / applicability filters	one at a time, paired + solitary	redundant but load-bearing	may not be recited as independently necessary
exclusion rule, conflict guard, authority, source status, recency, override	one at a time	essential	claim retained
ranking elements 1–3, specificity	one at a time	not measurable by that instrument	recorded NOT MEASURED, not "inert"; specificity claim withdrawn
global exclusivity	window decomposition	NEW → AMBIGUOUS flip	explains the identity veto

Table 5 — Blocked and not-run work, with exact unblock contracts

item	state	why	what would unblock it
Same-intelligence retrieval (W6)	not-run, closed	the confirmatory run was executed on 2026-08-20 and stopped before its arms. Acquisition succeeded (391 eligible / 359 excluded / 750 consumed / 0 unaccounted, corpus root frozen, development-title overlap 0) and produced 2,548 questions (185 revision-sensitive primary, 2,363 controls). The structural taint and residual leakage gates passed; the baseline validity gate refused on exact ties, 19/166 = 11.45% against a predeclared 10% ceiling. No arm ran; the comparative endpoint is NOT_MEASURED , not lost and not tied	nothing in this programme. The cohort is development data from this point forward, no threshold was relaxed, and no replacement benchmark is being built
Cross-source equivalence	not-run	offline survey found 0 explicit amendment relationships among 613 filing-shaped files	a second document family whose artifacts carry an explicit amendment or supersession pointer
Cause-conditioned recovery (A12)	blocked-external	no container runtime on the host; no build recipe for either candidate	an image build and publish producing a real registry digest
Experiment-time environment lock	open, mitigated	package digest is taken at manifest time	capturing the environment at execution time
Historical command recovery	partial	1 of 24 experiments has a recoverable original invocation	nothing — the other 23 are not recoverable and are labelled so
Build status under the project environment	passing	full suite 3,034 passed / 0 failed / 68 skipped under the project venv on 2026-08-20; 24 failures under the global interpreter were one environment defect, not a product defect	already discharged, but the figure is only meaningful with its interpreter, which the build-status receipt records

A correction that belongs in this table. The W6 row previously read "v5 acquisition worker died". That cause was inferred from an on-disk signature — one pair directory, no completion sidecar — and never verified. A second run reproduced the signature byte-for-byte and produced a traceback: a deterministic `ValueError` raised while computing a path for the receipt, on the first eligible page of every run. The blocker was a call-site defect, not an unreliable worker and not an external resource. It is stated here because the wrong cause had been carried in four documents, and because "the worker died" is the kind of explanation that closes an investigation instead of opening one.

None of these is presented as a pending success. Each is a state with a contract attached, and three of the five cannot be discharged from this environment at all.